

Health-Optimal Routing In Pedestrian Navigation Services

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ABSTRACT

People use various criteria for choosing routes, which may vary depending on location and time, purpose of trip, and personal preferences. Common routing criteria supported by current navigation services include shortest, fastest, least traffic, and least expensive (e.g., less fuel cost, toll free). While each optimal route is computed by using one of these criteria, there is currently no criterion that can be used to compute routes that are health-optimal. In this paper, we focus on a new routing criterion to compute health-optimal routes with the main objective of increasing physical activity. Those who are physically capable and motivated to walk can adapt a lifestyle that includes walking as a means to mitigate or prevent obesity. To that end, a routing criterion for computing health-optimal routes suitable for those who are concerned with obesity must take into account both environmental and individual factors. Computing optimal routes requires that each road segment of a road network be assigned a weight; like, distance for shortest routes and travel time for fastest routes. In this paper, we present and discuss a new weight for segments of pedestrian paths used in pedestrian navigation services to compute health-optimal routes. While health-optimal routes may address various health conditions, the objective of this work is to provide options for walking routes to increase regular physical activity as one means to help mitigate or prevent obesity. Weights are calculated by considering both environmental and individual parameters. The optimal-health weight is simulated using various scenarios. The results of the simulations show that the computed weights can be used to find health-optimal routes that are meaningful and consistent with walkability and obesity standards.

Categories and Subject Descriptors

D.3.3 [Information Sciences]: Geoinformatics.

General Terms

Algorithms, Design, Experimentation, Human Factors.

Keywords

Pedestrian Navigation: Physical Activity: Health-Optimal Navigation: Optimal-Health Routes.

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1. INTRODUCTION

Obesity has been a very important public health issue in the world; especially in the USA. More than one-third of U.S. adults (35.7%) and approximately 17% (or 12.5 million) of children and adolescents aged 2-19 years are obese [1]. Obesity is a risk factor for some of the leading causes of death from chronic disease and is considered by the World Health Organization to be a global epidemic. Obesity begets many other life threatening diseases like heart disease, stroke, type 2 diabetes and certain types of cancer [2]. Decreasing the prevalence of obesity requires making healthy choices, which includes regular physical activity. However, doing moderate or intensive physical activity regularly may not be possible for everybody given their time constraints, physical conditions, and other individual factors. But, those who are capable of walking may use this mode of activity to fight obesity with minimal impact on their other regular activities. To that end, a suitable routing technique can help individuals to plan manageable walking routes given both environmental constraints and individual preferences. For example, a person having BMI, (weight in kg)/(height in meters)², in the over-weight range with his/her personalized factors such as time constraint, walking speed, and amount of calories to burn, and environmental factors such as walkability score and route complexity may be recommended a different route relative to the shortest or the fastest route between a pair of origin and destination locations. Despite many improvements on route choices, current navigation services do not provide health-optimal routes. We believe that pedestrian navigation services should be able to provide various personalized alternative routes, especially those pertaining health concerns, where they recommend health-optimal walking routes while considering other preferences.

Routing algorithms in car navigation services require a weight on each segment of roads in a road network [3]. For example, length (of each road segment) is used as a weight to compute the shortest route whereas time (to travel through each road segment) is used as a weight to compute the fastest route. Similarly, to compute health-optimal routes there is a need to assign a weight to each pedestrian path segment which will represent the impact of the segment on health. To model such a weight requires taking both environmental factors and individual factors into account. Environmental factors mainly determine how walkable each pedestrian path segment is and individual factors include health conditions, such as BMI, and preferences that affect health. Environmental factors considered in this work are *segment complexity*, *walkability*, *segment length*, *weather condition*, *day light*, and *safety*. Individual factors considered are *BMI*, *time constraint*, *walk speed*, and *amount of calorie that one intends to burn*. There could be dependencies and/or influences within and among the both type of factors. One approach to analyze

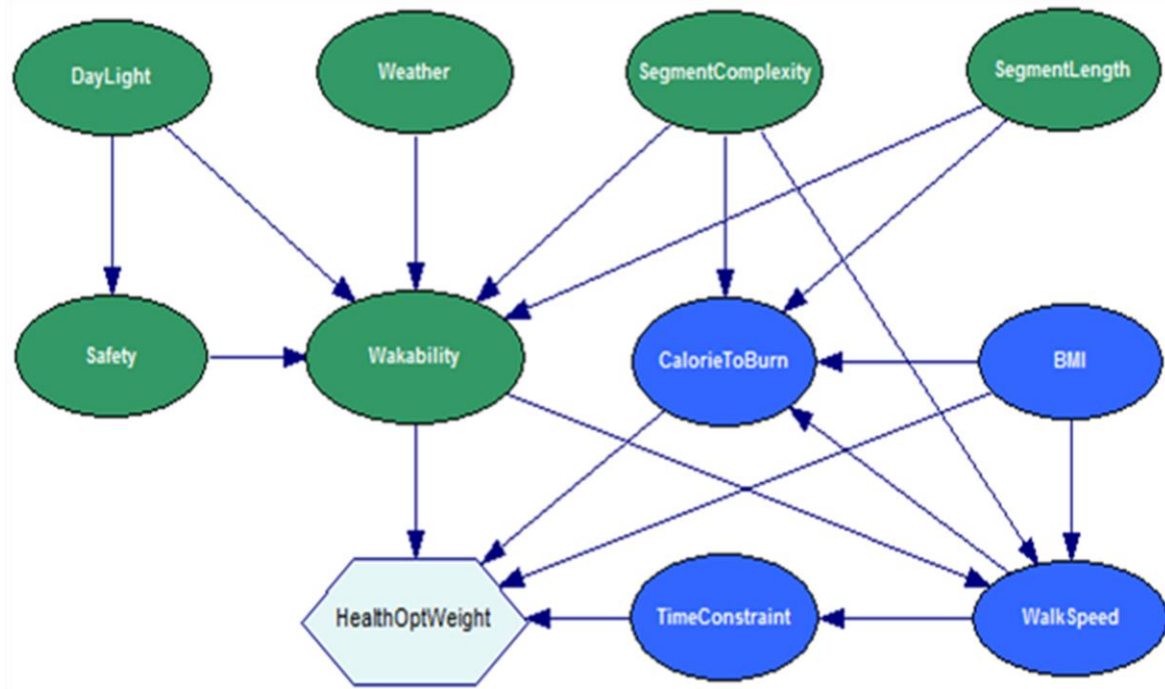


Figure 1. Influence diagram of health-optimal weighting model.

dependencies and influences within and among these environmental and individual factors is a Bayesian belief network (BBN). BBN represents a set of random variables, their conditional dependencies, and their probabilistic relationships by means of a directed acyclic graph [4]. The probabilistic relation and corresponding decision problem is represented and solved by influence diagram which is a generalized BBN. The theory behind influence diagram is based on Bayesian statistics. In this work, we developed a model considering both environmental and personal parameters and their influences on one another by using influence diagram. While BBN is not new, the idea of computing weights based on environmental and personal parameters to recommend health-optimal routes is novel and, to the best of our knowledge this is the first step toward developing more advanced models for health-optimal weights

In this paper, we present and discuss a health-optimal weighting model by taking environmental factors (green) and individual factors (blue) into account through an influence diagram as shown in Figure 1. The nodes of the graph represent the variables and the inter-dependencies among the variables are represented by the edges. Each of the variables, both environmental and individual, is assigned a probability for each of its states as defined in the influence diagram. The diamond symbol evaluates the expected utility score, which is in fact the weight for a pedestrian path segment, given all the possibilities of the variables. The model is designed such that the smaller the expected utility score, i.e., weight, for a pedestrian path segment, the healthier the segment will be. Once a weight is calculated for each pedestrian path segment and stored in a pedestrian path database, computing health-optimal routes for pairs of origin and destination locations will be straightforward. The contribution of the paper is the development of a model for calculating weights on pedestrian path segments that can be used to compute health-optimal routes. The health optimality in this paper is focused on mitigating

obesity by means of increasing physical activity. Though there are different physical conditions related to obesity, we considered BMI, which is an important indicator of obesity, in this work. The rest of the paper is organized as follows. Section 2 gives background to the work. Section 3 explains the weighting model in detail. Section 4 discusses simulation of the model followed by the results and discussion in Section 5. Section 6 summarizes the paper and provides some future work directions.

2. BACKGROUND

Walking is considered as one of the most natural physical activities. Walkability of a walking trail depends on many factors, both environmental and walker's preferences. Different factors such as distance from nearest points of interest (POIs), e.g., pharmacy, grocery, or theatre, as well as walking trail complexities (slope, up-hill, down-hill) can influence physical activity among children, adolescents and adults [5, 6]. The feasibility features that promote various forms of physical activity (such as walking) are referred to as walkability and often include access to destinations such as retail stores, parks, and features such as street connectivity and sidewalk access [7]. To measure walkability, traditional metrics vary considerably, from self-reported information to systematic field observation, also known as environmental audits [8]. Systematic field observations are very laborious (time wise, logistically) and often require significant specialized training and are less popular for being very costly. Geographic information systems (GIS) data have been used to evaluate neighborhood walkability [9]. The caveat to using GIS data to measure walkability is that using GIS also requires specialized expertise and also can involve time-intensive tasks [10]. Additionally, differences between pedestrian navigation and car navigation, lack of appropriate pedestrian network for pedestrian navigation system are some key issues to consider.



Figure 2. Routes calculated by using road and pedestrian network [29].

2.1 Walkability and Physical Activity

While walkability score can be measured based on walkable amenities such as retail stores [11] and several community design features such as street connectivity [12], other perspectives can be used to measure it. Weather condition, complexity of the walking trail, distance between origin and destination, time of the day, and safety could also be used to measure walkability score. Research examining walkability score and its validity has been conducted in local areas (in a neighborhood) which may not accurately represent the score for each pedestrian path segment of the area. Also, as it is possible that the validity of Walk Score varies by geographic location, generalization of the current research is limited. Current research suggests low correlations between objective and perceived neighborhood features [13]. In addition, current research on walkability score has been evaluated by comparing it to GIS-driven walkability indicators based on a 1-mile buffer only [11, 12] where location can be entered as geographic coordinates or as an address which is then geocoded using Google Geolocation [14]. Walkability measuring algorithms divide facilities into five categories—educational (e.g., schools), retail (e.g., grocery, pharmacies, convenience, bookstores), food (e.g., restaurants), recreational (e.g., parks, gyms) and entertainment (e.g., movie theaters)—and then calculate the closest straight-line distances to these facilities and a linear combination of these distances weighted both by facility type priority and a distance decay function [9]. Many of the destinations used in the Walk Score algorithm have been found effective to predict walking for exercise [15].

The relationships between environmental attributes and physical activity may be moderated by socio-demographic factors which could either influence the relationships between the environmental attributes and cognitions, or the relationships between such cognitions and physical activity [16]. There is a consistent body of evidence on the health benefits of physical activity [17, 18]. Over the last decade, there has been an increasing interest in finding the relationship between the physical environment as a determinant of physical activity, neighborhood built-environment attributes have been shown to be associated with walking [19, 20, 21], and obesity [22, 23, 24, 25]. According to ecological models of health behavior, appropriate opportunities and settings that facilitate particular forms of activity, such as walking for recreation and exercise, or walking to get to and from places, help adults to

achieve sufficient levels of physical activity for health benefits [26, 27]. However, there is a void in the literature on a weighting model for pedestrian path segments based on health issue that can help recommend health-optimal routes for given environment factors and individual constraints in pedestrian navigation services.

2.2 Pedestrian Navigation

As the health-optimal weight is suitable for walking, it is important to understand the differences between pedestrian navigation and car navigation. Pedestrian navigation is different than car navigation in many respects. Pedestrians walk along pedestrian paths, not along street lanes, and are not constrained by the boundaries of roads as cars are. Road networks used for car navigation are usually based on road centerlines whereas pedestrians travel along pedestrian paths such as sidewalk, crosswalk, walking trail, entrance to POI, pedestrian bridge, pedestrian tunnel [28, 29]. In the absence of pedestrian paths in some places, some navigation services, originally designed for guiding drivers, provide navigation assistance to pedestrians using road networks. In such services, the outcome is often not effective since pedestrian paths are usually on both sides of road centerlines. Since pedestrian paths could be of different types (e.g., sidewalk), road networks cannot be used in place of pedestrian paths for pedestrian navigation. Using road networks for guiding pedestrians will result in inaccurate, ineffective, and sometimes unusable outcomes. Another issue with the use of road networks for guiding pedestrians is related to origin and destination locations which may not be along road segments. To demonstrate the consequence of using a road network instead of a pedestrian network while computing routes for pedestrian, an experiment was conducted by [29] which computed shortest routes between three pairs of sources and destinations using both road network (derived from NAVTEQ) and pedestrian network developed by [30]. The result (Figure 2) shows that, using road networks, computed routes (the solid lines) for pedestrians will start and end near road segments, not at their exact locations. Furthermore, routes (the dotted lines) computed based on pedestrian paths will contain fine walking details between origin and destination locations which are unavailable in road networks. For instance, the dotted line between Pair 2 starts right at the origin point and arrives right at the destination point, while the solid line does not and is shorter.

Differences between road and pedestrian networks influence walkability index as experimented by using the Pedestrian Road Distance (PRD) method [31]. PRD is the ratio between the actual route distance travelled and the Euclidean distance between specific origins and destinations. The lower the PRD value the better the connectivity and hence the better the walkability index. PRDs are usually smaller on pedestrian paths compared to PRDs on road network. A challenge in pedestrian navigation is that GPS-based services are susceptible to accuracy degradation or signal blockage from satellite due to obstacles like trees and high rise buildings. Also, due to accuracy range of GPS receivers on mobile devices (e.g. smartphones), which is around 10 m, and the fact that the speed of movement by pedestrians, which is about 1 m/sec, is much slower than speed of movement by cars, GPS-based pedestrian navigation services update new positions at a longer time interval (~10 seconds). Furthermore, GPS signal is not always available on pedestrian paths near large trees, high-rise buildings, hills, and in tunnels.

Even if GPS signal is available, signal quality as well as accuracy may not be good enough for locating pedestrian accurately on the pedestrian network. Another limitation of GPS-based navigation services is that they cannot distinguish between the two sides of a street when the distance between the two sides is less than the positional accuracy range of the GPS unit [32]. In addition, pedestrian networks, in general, are denser than road networks within the same area. For these reasons, finding the correct sidewalk segment, on which the user travels, is a challenge.

3. WEIGHTING MODEL

An influence diagram and Bayesian statistics are used for the weighting model in this paper. The variables used in the model in each category, environmental and individual, are defined in Table 1. Each variable has different probabilistic states, and based on their probabilities, influence of a variable is determined over other variable(s). The overall influence determines the final weight.

3.1 Influence Diagram

An influence diagram [33] is a generalized BBN that gives a simple visual representation of a decision problem by means of an acyclic directed graph. It helps decision makers with an understanding of the results of actions they may consider to implement. The more insight details of the problem that the decision maker can incorporate into this tool, the more optimal results the tool will provide. In one BBN, the result may vary based on the input belief that it gets from different decision makers. Influence diagrams are constructed using four types of nodes (variables): *Decision*, *Chance*, *Deterministic*, and *Value* and two types of arcs: *influences* and *informational*.

Decision nodes (rectangles) represent variables that the decision maker can control and model the decision alternatives that the decision maker may have. Decision nodes also contain specifications of the available decision options. *Chance nodes* (circles or ovals) are random chance variables that represent uncertain quantities (quantified by conditional probability distributions) relevant to the decision problem. *Deterministic nodes* (double-circles or double-ovals) represent either constant values or algebraically determined values from the states of their parent nodes. If the value of any parent node is known then the value of its deterministic child node is also known with certainty. *Value node* (diamonds) represent a measure of desirability (utility) of the outcomes of the decision process based on possible influences from its parent nodes. An *influence arc* represents influence of one node on other nodes. The node at the tail of an arc influences the value (or the probability distribution over the possible values) of the node at the head of the arc. An arc from a decision node to a chance node denotes the impact (which is basically modifies the probability distribution of the child node) of decision nodes on chance nodes. But an arc incoming to a decision node gives a temporal precedence between the decision node and its parent. If there are multiple decision nodes in a diagram, *informational arcs* are used to connect the decision nodes so that one decision node is aware of the others.

Table 1. Description of influencing variables for health-optimal weighting model

Category	Variables	Description	States
Environmental	SegmentComplexity	Characterizes a segment based on its complexities such slope, uphill, downhill etc.	High, Average, Plain
	Walkability	Walkability scores are calculated based on local facilities, neighborhood type, road conditions etc.	High, Medium, Low
	SegmentLength	The shortest routing distance between start and end node of a segment	Long, Medium, Short
	Weather	Weather condition for the day	Good, Fair, Bad
	DayLight	Is the walk in day time or in night time?	Yes, No
	Safety	How safe the road to walk is. Safety is in terms of crime.	Safe, Not Safe
Individual	BMI	BMI, (weight in kg)/(height in meters) ² , is the most important parameter here to consider. Obesity is defined mainly by BMI and the Center for Disease Control (CDC) standard is followed.	Obese, Overweight(OW), Normal, Underweight(UW)
	TimeConstraint	If there any time constraint for an individual to reach to destination.	Flexible, Constraint
	WalkSpeed	Walking Speed of individual	High, Medium, Low
	CalorieToBurn	The amount of calorie individual wants to burn through walking	High, Medium, Low

3.2 Bayesian Modeling

BBNs represent the full joint distribution over the variables with smaller number of parameters. This takes advantages of conditional and marginal independences among random variables. For example, if variable A and variable B are independent then the joint probability is:

$$P(A, B) = P(A)P(B) \quad (1)$$

But if A and B are conditionally independent given variable C then:

$$P(A, B | C) = P(A | C)P(B | C) \quad (2)$$

$$P(A | C, B) = P(A | C) \quad (3)$$

From Bayes rule:

$$\begin{aligned} P(A | B) &= \frac{P(A, B)}{P(B)} \\ \Rightarrow P(A, B) &= P(A | B)P(B) \end{aligned} \quad (4)$$

The joint probability can be expressed in terms of conditional probability. So for walkability, the joint probability can be calculated as follows:

$$\begin{aligned} &P(w, sf, dl, wr, sc, sl) = \\ &P(w | sf, dl, wr, sc, sl)P(sf | dl)P(dl)P(wr)P(sc)P(sl) \\ \Rightarrow P(w | sf, dl, wr, sc, sl) &= \\ &\frac{P(w, sf, dl, wr, sc, sl)}{P(sf | dl)P(dl)P(wr)P(sc)P(sl)} \end{aligned} \quad (5)$$

where w is walkability, sf is safety, dl is day light, wr is weather, sc is segment complexity, and sl is segment length.

Similarly, for CalorieToBurn:

$$P(c | b, sc, sl, ws) = \frac{P(c, b, sc, sl, ws)}{P(ws | b)P(b)P(sc)P(sl)} \quad (6)$$

where c is calorie to burn, b is BMI, sc is segment complexity, sl is segment length, ws is walk speed.

And for TimeConstraint,

$$P(tc | ws) = \frac{P(tc, ws)}{P(ws)} \quad (7)$$

where tc is time constraint and ws is walking speed.

3.3 Health-Optimal Weight

A probability function is associated with each node in the influence diagram where the function takes a particular set of values for the node's parent variables as input and gives the probability of the variable represented by the node. For example, if there are n Boolean parent variables for a node x , then the probability function for x could be represented by a table of 2^n entries of true or false. If node x is also a Boolean variable, its values (2 states) are determined by the combination of its parent variables. It can be seen from the influence diagram in Figure 1 and the definition of variables in Table 1 that not all variables are Boolean. In these cases, a combination of all possible states determines the influence on the child node. For example, walkability node is influenced by five parent nodes, DayLight,

Safety, Weather, SegmentComplexity, and SegmentLength. From Table 1 we see that DayLight has 2 states, Safety has 2 states, Weather has 3 states, SegmentComplexity has 3 states, and SegmentLength has 3 states. So the overall influence on walkability is a product ($2 \times 2 \times 3 \times 3 \times 3$) of all possible state values. The weight for a pedestrian path segment, given individual's preferences and walkability of the path segment, incorporates all the probabilistic influences on it as in Equation (8) using marginalization

$$wt = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^4 \sum_{l=1}^2 P_i(w) \times P_j(c) \times P_k(b) \times P_l(t) \times s \quad (8)$$

where, wt -Weight, w -Walkability, c -CalorieToBurn, b -BMI, t -TimeConstraint, s -ScaleFactor.

The ScaleFactor is an assigned possible value for a given scenario that is believed to be realistic. In BBN, the beliefs could be set once and based on different states of the influencing variables, the scale value will be modified. The range of the ScaleFactor does not matter since the final weight will also be proportionally scaled.

4. SIMULATION

Currently no benchmark model is available for evaluation and/or verification of the developed model. Since the idea of weight for health-optimal routing is new, we analyze the model through simulations using the influence diagram. Various possibilities of constituent variables are set in the simulated environment. We ran the simulation for different scenarios to determine weights for all possible states of the variables for a given (under one belief) range of *ScaleFactors*. Setup for the simulation, and hence the values of chance variables, may vary depending on beliefs of decision maker about the initial values. An example scenario of a final weight for a pedestrian path segment given the probabilities of states for all the variables is shown in Figure 3 which is a bar chart view of the influence diagram. Here, all the values at different states of each chance variable are initialized as follows.

Under environmental variables, DayLight has two states Yes=50%, No=50% which means user does not care about day time or night time walking. Weather has three states Good=50%, Fair=30%, Bad=20% which means the user is considering the weather condition is better than average. States of the SegmentComplexity are set as Plain=20%, Average=30%, and High=50% which consider the path segment complexity as above average. SegmentLength states are set as Short=10%, Medium(Mid)= 30%, and Long=60% denoting the segment in consideration is longer than the average segment length. For Safety, states are Safe=50%, Not Safe=50% which again means that the user does not care about the safety while walking. Finally Walkability states are set based on the influences of other influencing variables (DayLight, Weather, SegmentComplexity, SegmentLength, and Safety) on it. For each of the 108 different scenarios (for DayLight 2 states, for Weather 3 states, for SegmentComplexity 3 states, for SegmentLength 3 states, and for Safety 2 states, i.e. $2 \times 3 \times 3 \times 3 \times 2 = 108$ combinations of states for five influencing variables) one set of values for Walkability states are initialized.

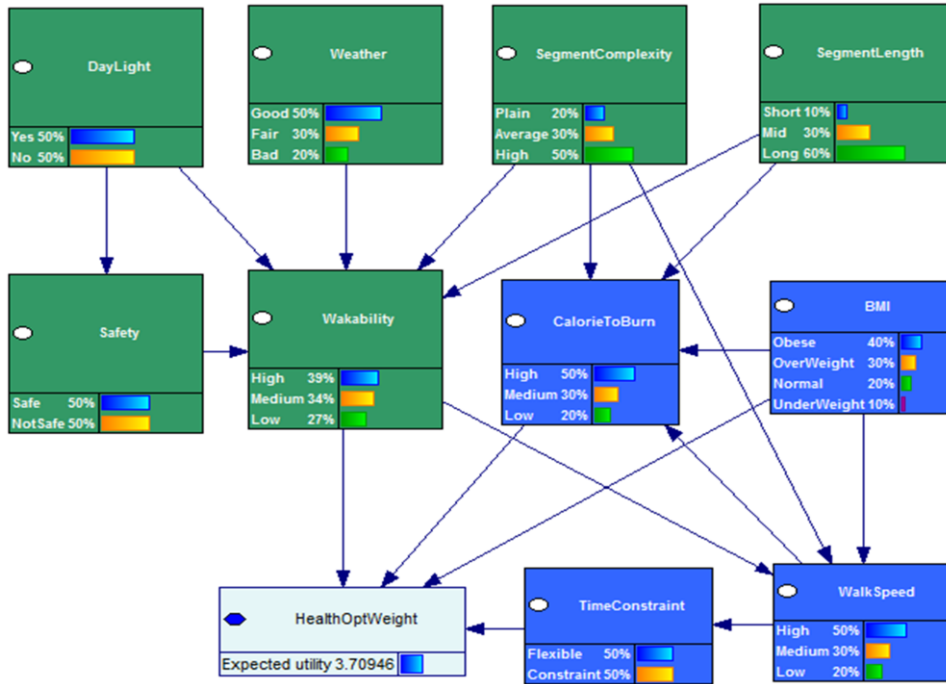


Figure 3. A simulated instance of health-optimal weighting model.

Similarly, for personal factors and preferences, BMI states are set as Obese=40%, Overweight=30%, Normal=20%, and Underweight=10% which means that the weight will be calculated for different such possibilities of BMI. CalorieToBurn states are initialized as High=50%, Medium=30%, and Low=20% which means that the user wants to burn more calorie than average. States for TimeConstraint are set as Flexible=50% and Constraint=50% denoting no special care is to be taken on time constraint. Finally, WalkSpeed states are set as High=50%, Medium=30%, and Low=20% which again means that the weight is expected for a faster walking phase. It is worth mentioning here that all of these initialized values are set based on an individual belief at a particular instant.

In this example, the walkability is calculated based on five influencing parent (influencing) variables. A probabilistic function is involved with each variable that determines its influences on its child variables as well as the influence it gets from its parent nodes. For instance, the influence of five environmental variables on walkability follows Equation (5) where CalorieToBurn gets influences on it following Equation (6) and TimeConstraint experiences influence on it as followed by Equation (7). One important individual factor is BMI which influences WalkSpeed, CalorieToBurn, and finally the weight itself.

On the other hand, CalorieToBurn is influenced by BMI, WalkSpeed, SegmentComplexity, and SegmentLength but it directly influences the final weight. This is just one scenario of all simulations. Other simulations were conducted by varying the probabilities of variable states for each variable. We examined the changes in weight generation for the four variables (BMI, Walkability, CalorieToBurn, and TimeConstraint) that directly influence the weight by using Equation (8). Also, the weight is generated for each given states of each of the directly influencing variables. For example, it is examined how the overall weight is

influenced for walkability when it is set to high in comparison to the weight calculated for different probabilities set to different walkability states. The final weight will be in the range of 0 to 10 where the lower weight denotes the higher health-optimality.

5. RESULTS AND DISCUSSION

Many different probability values could be initialized for the chance variables for many different beliefs. This paper demonstrates the results for each of the directly influencing variable separately on final weight. The results are shown (Figures 4-8) in terms of calculated weights (upward diagonal bars) for and impacts (solid bars) by path segments. Impact actually shows how effective the weight would be while computing a health-optimal route. It is used for visualizing the effect of weights. The lower the weight, the higher impact as health optimality is experienced. For the example scenario setup as shown in Figure 3, the combined influences of the five environmental variables (DayLight, Weather, SegmentComplexity, SegmentLength, and Safety) on Walkability is found as High=39%, Medium=24%, and Low=27%. Note that these percentages are set based on the equivalent probabilities for states of each variable. This means that based on the believed probabilities for different states of the influencing variables, as set for all 108 different combinations, the overall influence on the walkability results 39%, 24%, and 27% influence by High, Medium, and Low walkability, respectively, on the final weight calculation. In other words, the ScaleFactor value is affected from Walkability consideration (high, medium, and low) as 39, 24, and 27 proportions. Similarly, the ScaleFactor is affected by the influences of BMI, CalorieToBurn, and TimeConstraints in different proportion resulting in the overall expected utility score of 3.71. This score is treated as the computed weight for the pedestrian path segment in consideration. Since the weight value ranges between 0 to 10, 3.71 means an above average health-optimality for the given scenario.

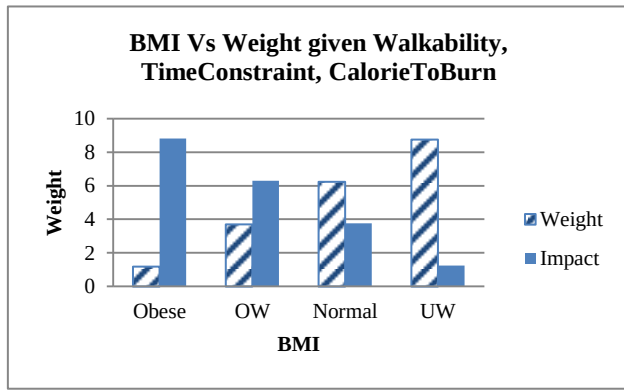


Figure 4. Weights/Impacts for different BMIs.

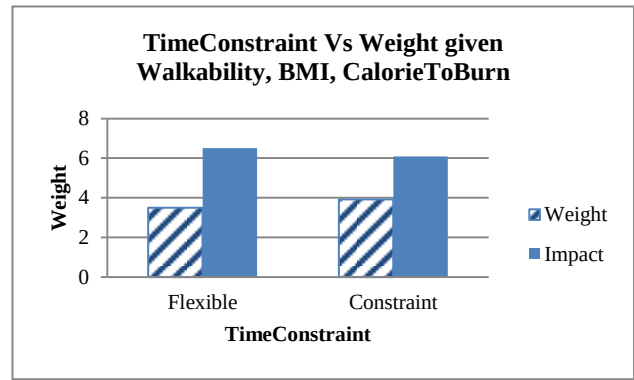


Figure 6. Weights/Impacts based on time constraint.

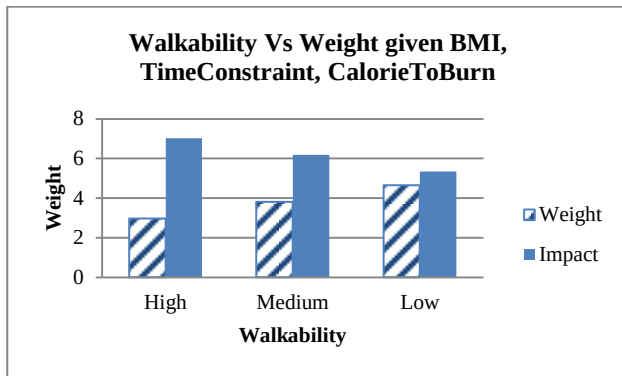


Figure 5. Weights/Impacts for different Walkability states.

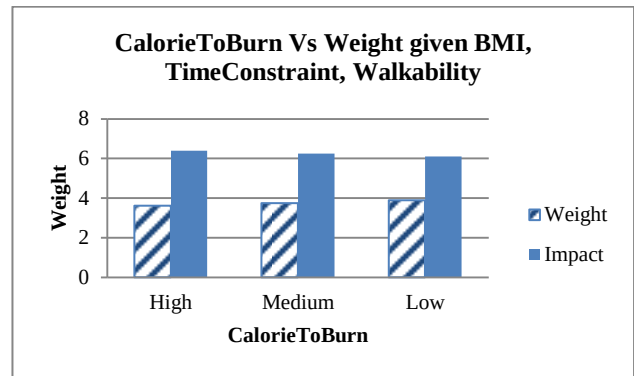


Figure 7. Weights/Impacts for calorie burn.

For BMI, the generated weight for a path segment given all other variable states is shown in Figure 4. The result shows that for the same path segment the calculated weight is different for persons with different BMI. For example, the weight is lower for an overweight person than a person who is underweight. This means that the segment is more health-optimal for the overweight person than it is for an underweight person. Similarly, as shown in Figure 5, if the walkability of a path segment is high, its weight is low which validates the idea of optimality (the lower the better) with respect to health.

On the other hand, the weight for situations with flexible time is smaller than the weight for situations with time constraint (Figure 6). This means that there would be more alternative health-optimal routes when reaching destination where time is not an issue compared with when there is a time constraint to reach destination. Another way to infer this is that a different health-optimal route would be recommended to an individual who wants to burn more calories compared to an individual who is less concerned about burning calories (Figure 7).

One of the special cases is examined and simulated as shown in Figure 8. In this case, the walkability is set to high meaning that one is sure about the walkability status(i.e. highly walkable) for the path segment. Now the weight generated for this status change is 0.448 for obese person where in the previous case (Figure 4) the weight for obese is 1.19. That means, a 100% walkable segment is more health-optimal than a segment with walkability less than 100%. This particular simulation result is shown in Figure 9. Similarly, the same path segment is found more health-optimal for

one with BMI in obese range than that for one with BMI in overweight range.

To demonstrate the individual effect of each of the variables that directly influence the expected utility, the final weight is computed and stratified over the influencing variables. In reality, all the influences of individual variables, based on initial probability and belief, are applied together on the final expected utility which is the final weight. The results obtained in this simulation may be a partial representation of a real world scenario since more potential personal factors and environmental factors could be incorporated into the model. However, as a first attempt, the factors help to have a clear idea about the model. Consideration of other factors may increase the possibility of obtaining more accurate weights and consequently routes that address health at individual level. For example, in the current version, the model will not provide a route for a requested time or distance; rather it will generate weights for road segments based on probabilities set for TimeConstraint and SegmentLength factors.

6. SUMMARY AND FUTURE WORK

This work is a demonstration of a model which takes some environmental and some personal variables into account to compute health-optimal weights for segments of a pedestrian path. Each variable is assigned values for different states, representing user beliefs using influence diagram which is a generalized BBN, and influences of different variables on other variables as well as

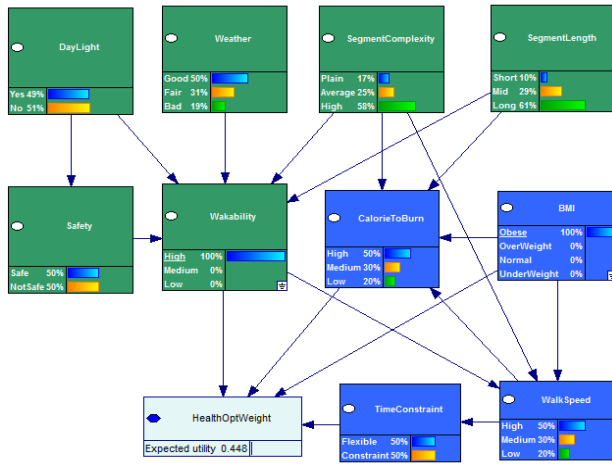


Figure 8. A health-optimal weighting model for obese and for segment walkability=high.

on the weight to calculate the overall weight. The weights can then be used to compute health-optimal routes based on the states of both the environmental and personal variables. The optimality will vary for individuals based on the probabilities of states of each individual variable. This is reasonable because the probabilistic belief for each variable is different for different individuals. In this paper, we used a BBN model and took a forward approach, i.e. we constructed the network, determined dependencies, assigned values based on belief, and calculated the weight by means of utility score. The variables and their values were simulated and the results were analyzed and discussed.

The work presented in this paper is considered as a foundation for health-optimal routing in pedestrian navigation services. Future research includes:

- Investigate additional parameters for both environmental factors and personal factors toward developing personalized routes with the objective of mitigating and preventing obesity.

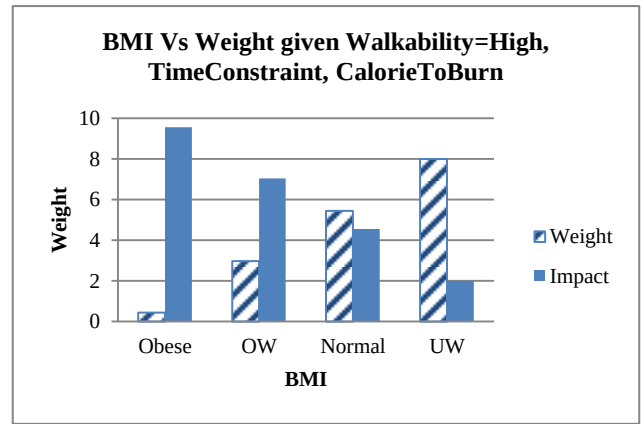


Figure 9. Weights/Impacts for different BMIs for given walkability (=high).

- Investigate inclusion of new variables that impact other health issues like heart conditions and diabetes. Also, more environmental variables like traffic and pollution could be included to recommend more effective health-optimal routes.
- Investigate the weighting and impacts using a backward approach in BBN, i.e., constructing the network from data, provided suitable data is available.
- Investigate other techniques for developing health-optimal weights and compare them with the current technique of BBN.
- Compute optimal-health routes by assigning health-optimal weight to each segment of a pedestrian path database, and compare them with alternative routes, e.g., shortest routes.
- Perform subject testing to validate the results obtained by simulation
- Incorporate the proposed health-optimal routing technique into a pedestrian navigation service.

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